Day 16: Wazuh Lab Deployment Preparation

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Date	19 September 2026
Internship Track	SOC (Security Operations Center)

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Key Terms

- **Wazuh:** Open-source SIEM and XDR platform used to collect, correlate, and alert on security-relevant log data from monitored endpoints (agents).
- **SIEM / XDR:** Security Information and Event Management / Extended Detection and Response: platforms that centralize log collection, correlation, and alerting across an environment.
- **Hypervisor:** Software (here, VMware Workstation Pro) that creates and runs virtual machines on physical host hardware.
- **LVM (Logical Volume Manager):** A Linux storage abstraction layer that pools physical/virtual disk space into a Volume Group, from which Logical Volumes are carved out for filesystems.
- **Volume Group / Logical Volume:** A Volume Group is a pool of disk space; a Logical Volume is a resizable slice of that pool that a filesystem is placed on.
- **resize2fs:** Linux utility that grows or shrinks an ext4 filesystem to match the size of the underlying Logical Volume.
- **Bridged Networking:** A VM network mode that places the virtual machine directly on the physical LAN with its own IP address, as opposed to NAT, which isolates the VM behind the host on a private virtual subnet.
- **DHCP:** Dynamic Host Configuration Protocol: automatically assigns an IP address to a device on the network.
- **TUI Installer:** Text-based (keyboard-driven) user interface, used by the Ubuntu Server installer in place of a graphical, mouse-driven interface.

Executive Summary

This report documents the preparation of a dedicated Wazuh server virtual machine ahead of installation. The original lab host (a Dell Inspiron 15 3000 laptop, 8GB RAM, 221GB SSD) had only 21.6GB of free disk space after cleanup, insufficient for a Wazuh deployment alongside its existing Kali, Metasploitable, and Windows 11 lab VMs. The task was therefore moved to a second physical machine (a Dell PC with 8GB RAM and a 1TB HDD), and a new virtual machine, wazuh-server, was created on it using VMware Workstation Pro with Ubuntu Server 24.04.5 LTS, the newest Ubuntu release Wazuh officially supports.

The VM's network adapter was set to Bridged mode rather than the VMware default (NAT) so that wazuh-server sits on the same physical network segment as the existing lab VMs, which is required for those VMs to later be reachable as Wazuh monitoring agents. Following installation, a full verification pass confirmed supported OS, correct hostname, network connectivity, administrative access, and available RAM.

One issue was identified and corrected during verification: the Ubuntu installer's guided storage setup had allocated only 19GB of the 40GB virtual disk to the usable filesystem, leaving the remainder unassigned inside the LVM volume group. This was resolved using lvextend and resize2fs, bringing usable disk space to 38GB (29GB free). Two deliberate, documented compromises remain against Wazuh's official small-deployment recommendations: 6GB RAM allocated against an official minimum of 8GB, and 38GB usable disk against an official minimum of 50GB, both constrained by the host machine's own 8GB total RAM and by keeping the VM on a single virtual disk. The environment is otherwise fully prepared for Wazuh installation.

1. Objective

Prepare the Wazuh server virtual machine for installation by verifying its operating system is supported by Wazuh, confirming its hostname, network connectivity, disk space, and RAM are adequate, confirming administrative (root/sudo) access is available, and recording the final network plan before proceeding to the Wazuh installation itself.

In practical terms, this is the checkup a machine gets before software as significant as a SIEM is installed on it. Wazuh will end up handling security log data and, eventually, agent connections from other machines on the network, so it matters that the VM is running a version of Ubuntu Wazuh actually supports, that it can be found and reached on the network under a sensible name, that it has enough disk and memory to not fall over under load, and that whoever is installing Wazuh has the administrative rights to do so. Catching a problem with any of these now, before the Wazuh installer runs, is far easier than troubleshooting it halfway through an install or after Wazuh is already running.

2. Data, Environment, and Source Declaration

2.1 Environment

- Host machine: secondary Dell PC, 8GB RAM, 1TB HDD.
- Hypervisor: VMware Workstation Pro (free for personal, commercial, and educational use).
- Guest virtual machine: wazuh-server, Ubuntu Server 24.04.5 LTS (Noble Numbat), stored on host D: drive.
- Allocated resources: 2 virtual CPU cores, 6144MB (6GB) RAM, 40GB virtual disk (single file, split into multiple sub-files).
- Network adapter: Bridged.

2.2 Source Declaration

All command output, screenshots, and figures in this report were captured directly from the live wazuh-server VM during this session; no data in this report is simulated or synthetic. Wazuh's supported operating system list and official hardware recommendations were verified against Wazuh's current published documentation rather than assumed. All terminal commands were run interactively by the report author; none were pre-scripted or automated.

3. Deployment Decisions and Documented Compromises

3.1 Host machine change

The original lab laptop (Dell Inspiron 15 3000, AMD Ryzen, 8GB RAM, 221GB SSD, no secondary hard disk) had only 21.6GB of free storage remaining after cleanup, once its existing Kali Linux, Metasploitable, and Windows 11 virtual machines (roughly 65GB combined) and the host operating system (roughly 120GB) were accounted for. This was not enough headroom to safely create a new virtual machine sized for a Wazuh deployment. The task was therefore moved to a second available physical machine, a Dell PC with a 1TB HDD, which had substantially more free disk space (720GB on its D: drive) available for the new VM.

3.2 Bridged networking

The wazuh-server VM's network adapter was set to Bridged mode instead of VMware's default NAT mode. NAT would have placed wazuh-server on an isolated virtual subnet (typically 192.168.84.x) reachable only from its own host, cut off from the Kali, Metasploitable, and Windows 11 lab VMs running on the original laptop. Bridged mode instead places wazuh-server directly on the real home network (192.168.0.x), the same segment the laptop's lab VMs can also be placed on. This is a prerequisite for Wazuh to later receive agent traffic from those lab VMs across machines.

3.3 RAM allocation (documented compromise)

Wazuh's official minimum for a small all-in-one deployment (approximately 25 agents, 90 days of data retention) is 4 vCPU and 8GB RAM. The host Dell PC has only 8GB total RAM, all of which cannot be given to a single guest VM without starving the host operating system. 6GB RAM (2 vCPU) was allocated instead, below the official recommendation. This is accepted as a deliberate, documented lab-scale compromise, not an oversight.

3.4 Disk allocation (documented compromise)

Wazuh's official minimum disk recommendation for the same small all-in-one profile is 50GB. The wazuh-server VM was created with a 40GB virtual disk (38GB usable after filesystem formatting, further reduced from an initial 19GB before the LVM correction described in Section 5). This is below the official minimum and is accepted as a second deliberate, documented compromise, made to fit the constraints of the available hardware for this lab exercise.

4. Command Reference

The following commands were used to verify the prepared environment and to correct the disk allocation issue identified during verification.

4.1 Verification commands

```
cat /etc/os-release
hostname
ip a
ping -c 4 8.8.8.8
df -h
free -h
sudo whoami
```

- **cat /etc/os-release**: prints the installed OS name and version, used to confirm Ubuntu 24.04.5 LTS, a Wazuh-supported release.
- **hostname**: prints the machine's configured hostname, used to confirm it matches the intended name, wazuh-server.
- **ip a**: lists network interfaces and their assigned IP addresses, used to confirm the VM received a real address on the home network via Bridged mode.

- **ping -c 4 8.8.8.8**: sends four ICMP echo requests to a known public address, used to confirm outbound internet connectivity.
- **df -h**: reports filesystem disk usage in human-readable form, used to confirm available disk space.
- **free -h**: reports RAM and swap usage in human-readable form, used to confirm available memory.
- **sudo whoami**: runs the whoami command with elevated privileges, used to confirm administrative access is available (expected output: root).

4.2 Disk correction commands

```
sudo lvextend -l +100%FREE /dev/ubuntu-vg/ubuntu-lv
sudo resize2fs /dev/mapper/ubuntu--vg-ubuntu--lv
```

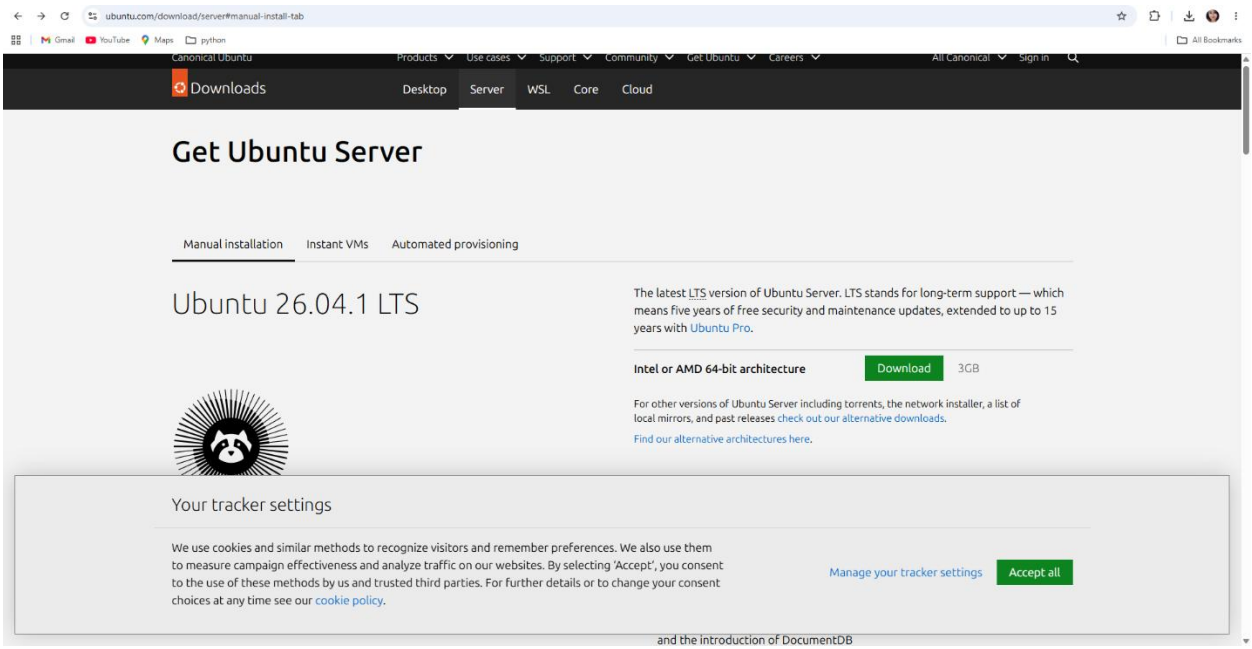
- **sudo lvextend -l +100%FREE /dev/ubuntu-vg/ubuntu-lv**: extends the Logical Volume ubuntu-lv to consume all remaining unallocated space in its Volume Group, growing it from 19GiB to 38GiB.
- **sudo resize2fs /dev/mapper/ubuntu--vg-ubuntu--lv**: grows the ext4 filesystem itself to fill the now-larger Logical Volume; performed online, with no reboot and no data loss, while the filesystem remained mounted.

5. Step-by-Step Execution

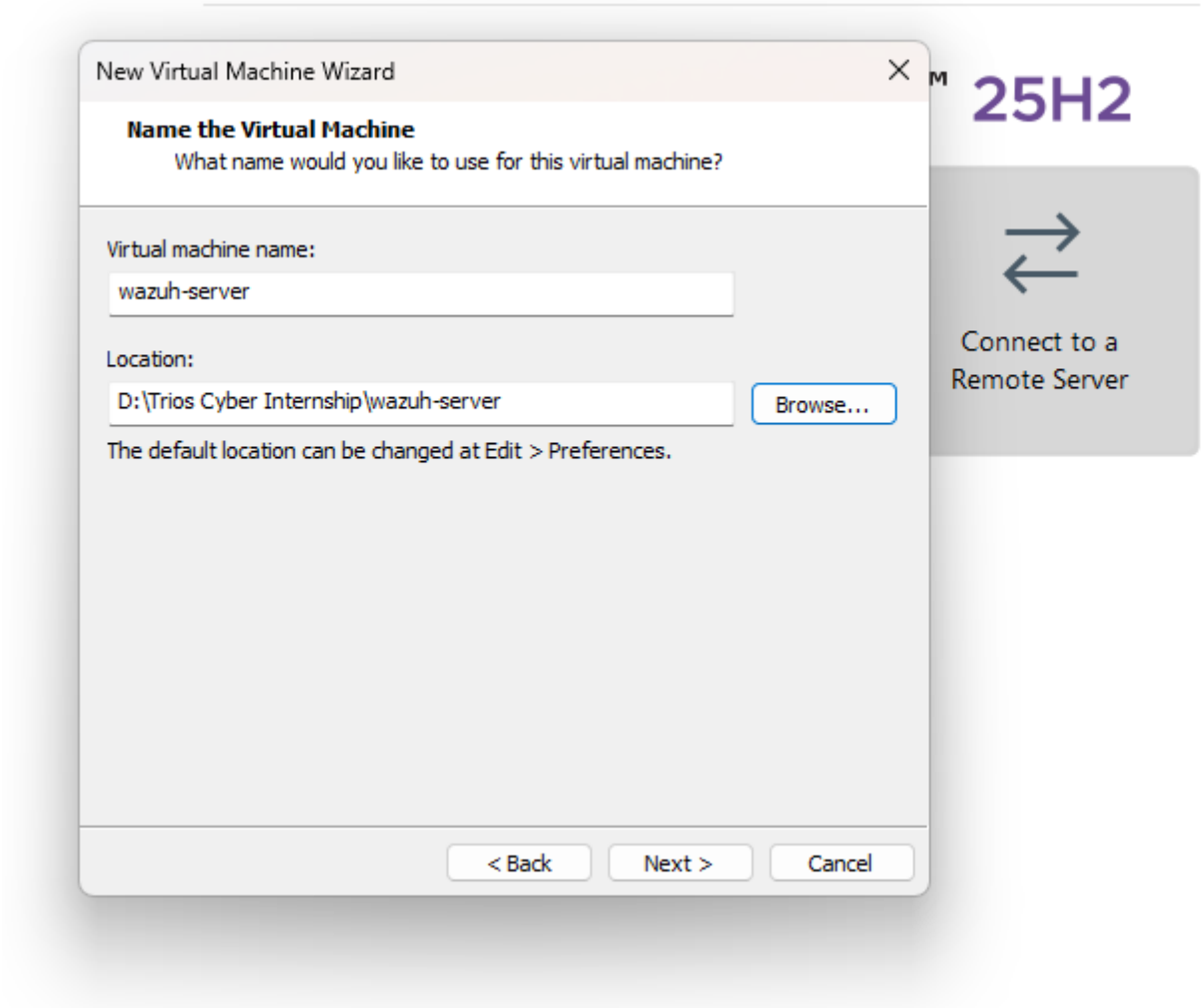
5.1 Virtual Machine Creation

Performed in VMware Workstation Pro on the host Dell PC.

Step 1 - Created a new VM, pointed to the downloaded Ubuntu Server 24.04.5 LTS ISO: VM set to boot directly into the Ubuntu Server installer.



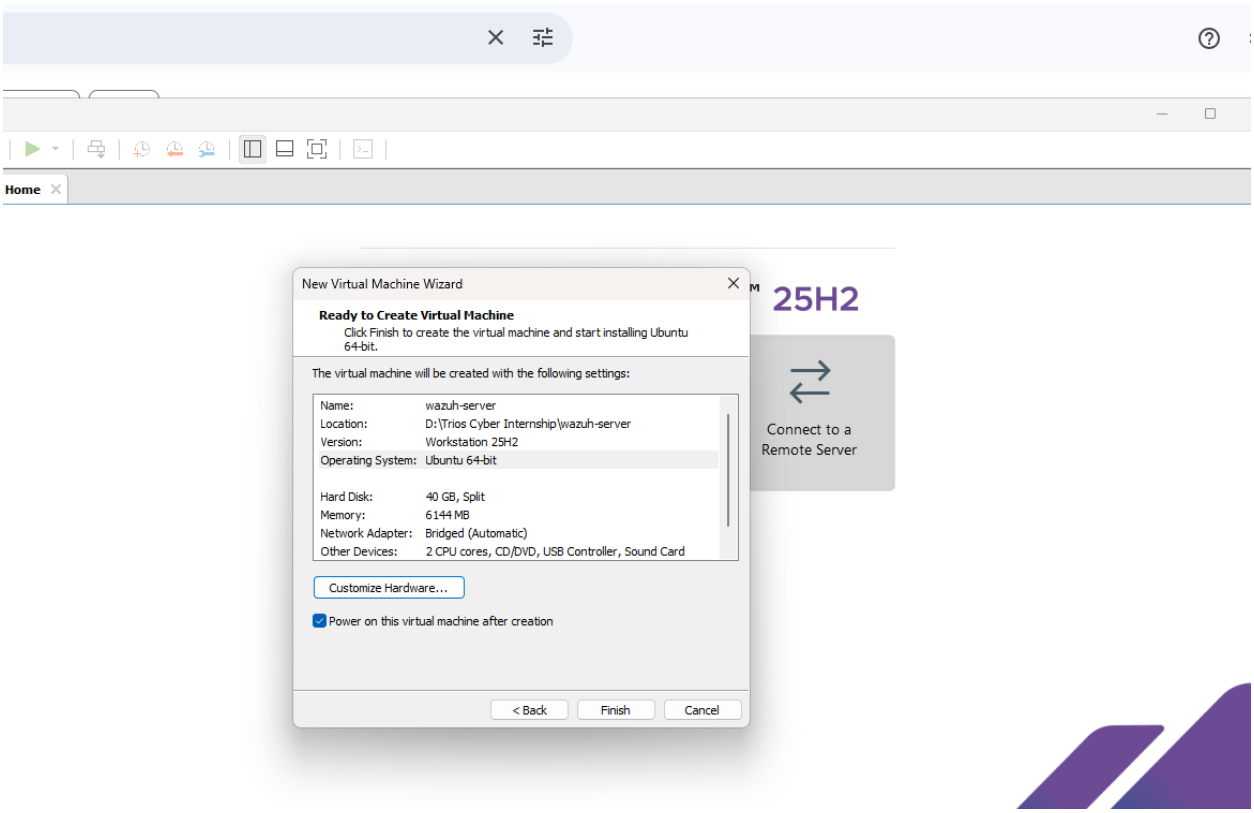
Step 2 - Named it wazuh-server, location D:\Trios Cyber Internship\wazuh-server: D: chosen for its 720GB free space (versus roughly 92GB on C:); named for its intended role.



Step 3 - Set disk capacity to 40GB, split into multiple files: Standard, portable disk format.

Step 4 - Customized hardware before finishing: Memory 6144MB (6GB), 2 processor cores, Network Adapter set to Bridged: RAM below Wazuh's 8GB recommendation (documented compromise, Section 3.3); Bridged chosen over NAT (Section 3.2).

Step 5 - Selected Finish and powered on the VM: wazuh-server booted into the Ubuntu Server installer.



5.2 Ubuntu Server 24.04 Installation

Completed inside the Ubuntu Server text-mode (TUI) installer, navigated with Tab, arrow keys, and Enter rather than the mouse.

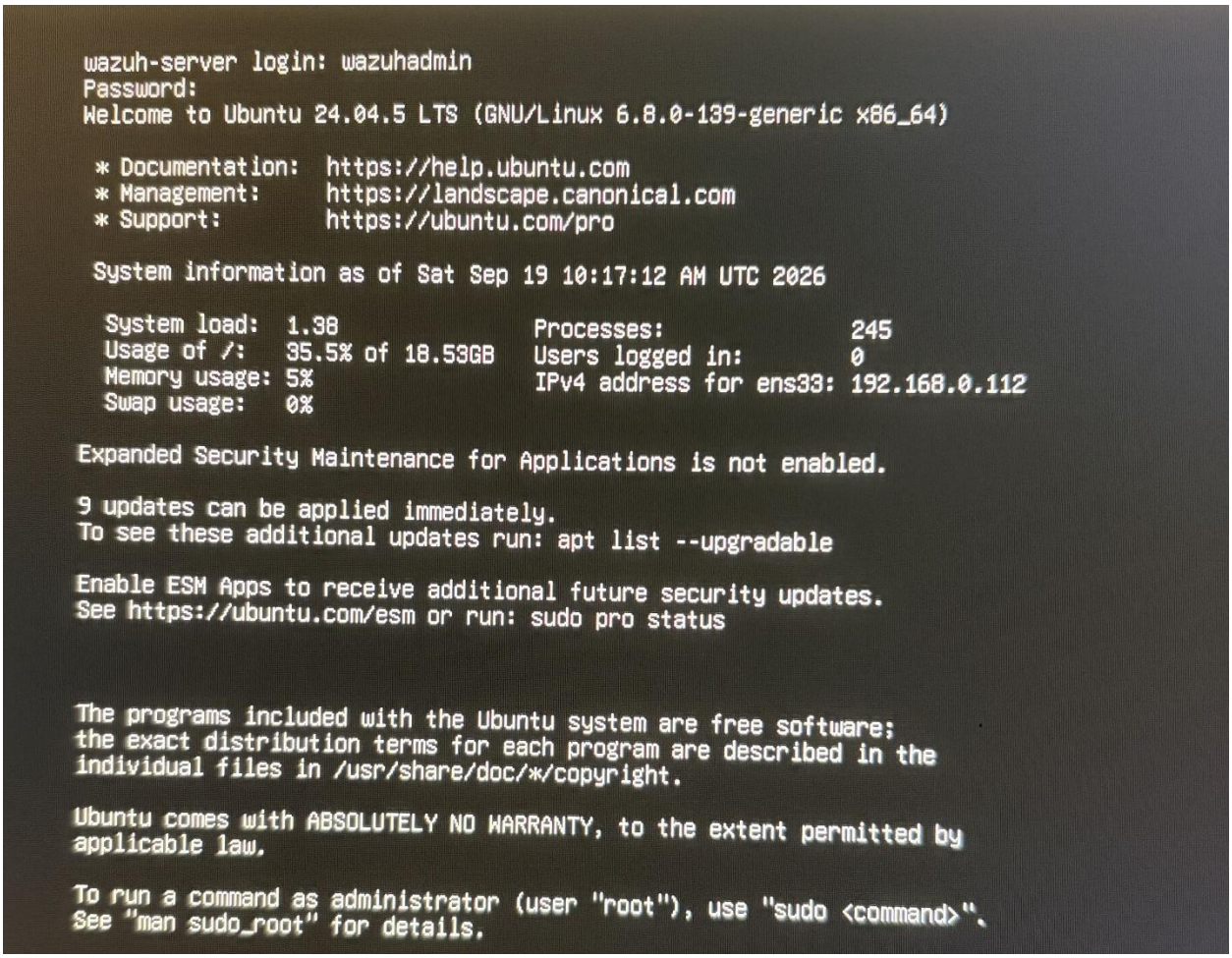
Step 1 - Selected language English and installation type Ubuntu Server: Not the minimized variant (risks missing dependencies Wazuh later expects) and not the Desktop variant (unnecessary GUI overhead).

Step 2 - Left networking on DHCP: Interface ens33 immediately received a live address, 192.168.0.111/24, confirming the Bridged adapter was working.

Step 3 - Left proxy and mirror settings at default: No proxy required on this network; mirror check completed automatically.

Step 4 - Selected "Use an entire disk" for storage and confirmed the destructive-action warning: Warning applies only to the new, empty 40GB virtual disk, not to any real hardware or other VM.

Step 5 - Set up the profile: hostname wazuh-server, username wazuhadmin, password: Matches this report's hostname requirement.



Step 6 - Selected "Install OpenSSH server"; skipped all optional Featured Server Snaps: OpenSSH enables pasting Day 17's Wazuh commands over SSH instead of typing them into the console.

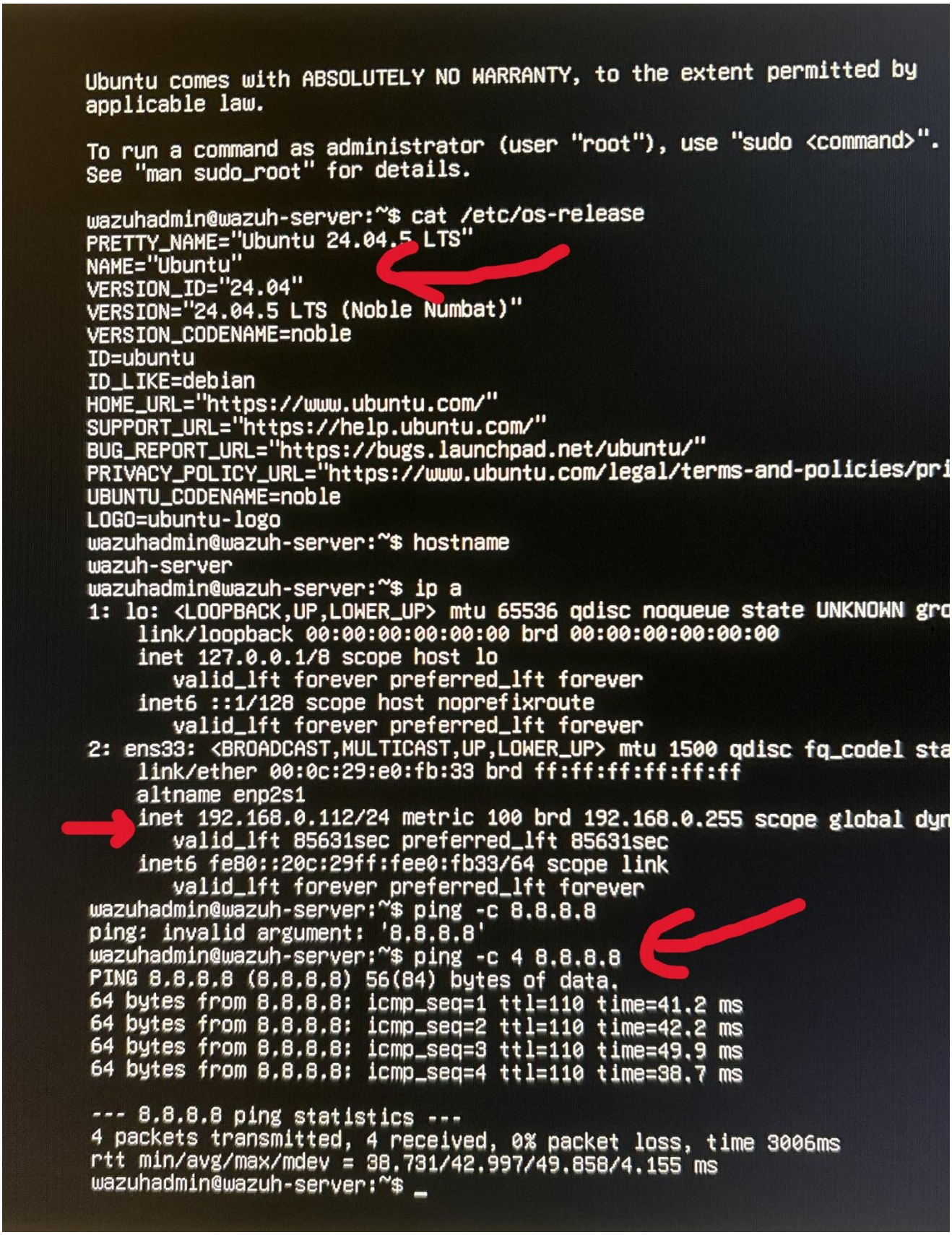
Step 7 - Let the installer run, then rebooted and logged in: Install ran slower than expected due to the host's HDD storage; a benign "failed unmounting cdrom.mount" message on reboot was cleared by pressing Enter.

5.3 Verification

Step 1 - `cat /etc/os-release`: PRETTY_NAME="Ubuntu 24.04.5 LTS" (Noble Numbat). Confirmed as a Wazuh-supported operating system.

Step 2 - `hostname`: wazuh-server. Matches the hostname set during installation.

Step 3 - `ip a`: Interface ens33 (altname enp2s1) shows inet 192.168.0.112/24, dynamic, scope global. Confirms Bridged mode placed the VM on the real home network.


Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by applicable law.
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.
wazuhadmin@wazuh-server:~$ cat /etc/os-release
PRETTY_NAME="Ubuntu 24.04.5 LTS"
NAME="Ubuntu"
VERSION_ID="24.04"
VERSION="24.04.5 LTS (Noble Numbat)"
VERSION_CODENAME=noble
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/pri
UBUNTU_CODENAME=noble
LOGO=ubuntu-logo
wazuhadmin@wazuh-server:~$ hostname
wazuh-server
wazuhadmin@wazuh-server:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN gro
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
valid_lft forever preferred_lft forever
inet6 ::1/128 scope host noprefixroute
valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel sta
link/ether 00:0c:29:e0:fb:33 brd ff:ff:ff:ff:ff:ff
altname enp2s1
inet 192.168.0.112/24 metric 100 brd 192.168.0.255 scope global dyn
valid_lft 85631sec preferred_lft 85631sec
inet6 fe80::20c:29ff:fee0:fb33/64 scope link
valid_lft forever preferred_lft forever
wazuhadmin@wazuh-server:~$ ping -c 8.8.8.8
ping: invalid argument: '8.8.8.8'
wazuhadmin@wazuh-server:~$ ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=110 time=41.2 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=110 time=42.2 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=110 time=49.9 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=110 time=38.7 ms
--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3006ms
rtt min/avg/max/mdev = 38.731/42.997/49.858/4.155 ms
wazuhadmin@wazuh-server:~$

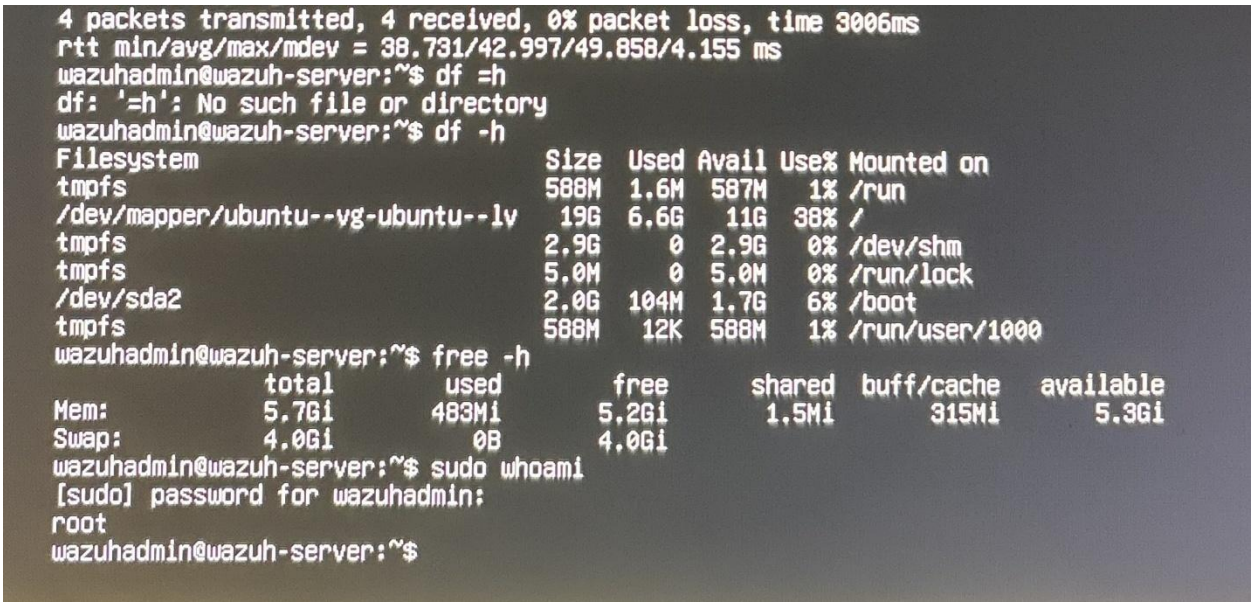
cat /etc/os-release, hostname, and ip a output on wazuh-server.

Step 4 - `ping -c 4 8.8.8.8`: 4 packets transmitted, 4 received, 0% packet loss, rtt avg 42.997ms. Confirms internet connectivity.

Step 5 - `df -h`: (initial run) Root filesystem (/dev/mapper/ubuntu--vg-ubuntu--lv) showed 19G total, 11G available. Below the intended 40GB allocation; investigated further.

Step 6 - `free -h`: Mem total 5.7Gi, available 5.3Gi. Consistent with the 6144MB (6GB) allocated to the VM, minus the small amount reserved by the kernel and firmware.

Step 7 - `sudo whoami`: root. Confirms administrative access via sudo.

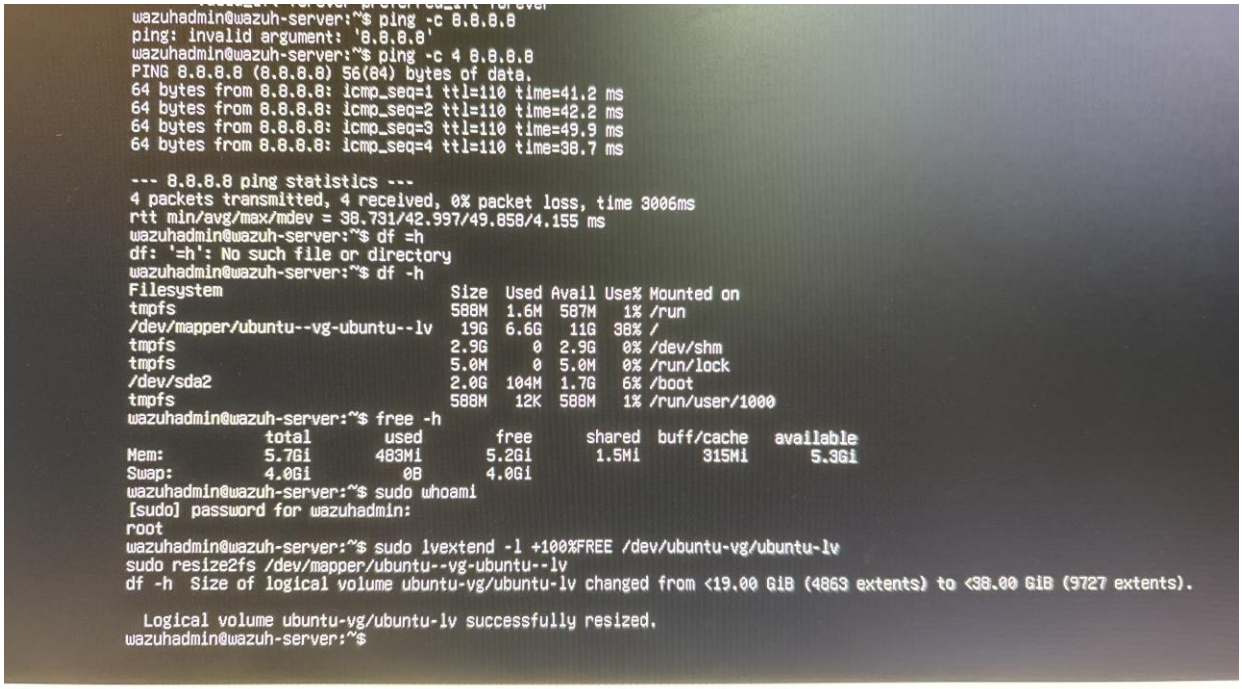


ping -c 4 8.8.8.8 output, 0% packet loss; df -h (initial, 19G total), free -h, and sudo whoami output.

5.4 Disk Correction

The initial df -h result showed only 19GB of the 40GB virtual disk in use by the root filesystem. This is a known behavior of Ubuntu's guided LVM partitioning: even when "use entire disk" is selected, the installer does not always assign all available space to the Logical Volume by default, leaving the remainder idle inside the Volume Group. The following steps corrected this.

Step 1 - `sudo lvextend -l +100%FREE /dev/ubuntu-vg/ubuntu-lv`: Size of logical volume ubuntu-vg/ubuntu-lv changed from 19.00 GiB (4863 extents) to 38.00 GiB (9727 extents). Logical volume successfully resized.



sudo lvextend output confirming the logical volume grew from 19GiB to 38GiB

Step 2 - `sudo resize2fs /dev/mapper/ubuntu--vg-ubuntu--lv`: Filesystem resized online (mounted on /). old_desc_blocks = 3, new_desc_blocks = 5. The filesystem on /dev/mapper/ubuntu--vg-ubuntu--lv is now 9960448 (4k) blocks long, approximately 38GB.

Step 3 - `df -h`: (final run) Root filesystem now shows 38G total, 6.6G used, 29G available, 19% used. Disk allocation issue resolved.

```
64 bytes from 8.8.8.8: icmp_seq=4 ttl=110 time=38.7 ms

--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3006ms
rtt min/avg/max/mdev = 38.731/42.997/49.858/4.155 ms
wazuhadmin@wazuh-server:~$ df =h
df: '=h': No such file or directory
wazuhadmin@wazuh-server:~$ df -h
Filesystem                         Size  Used Avail Use% Mounted on
tmpfs                              588M  1.6M  587M   1% /run
/dev/mapper/ubuntu--vg-ubuntu--lv   19G  6.6G   11G  38% /
tmpfs                              2.9G     0  2.9G   0% /dev/shm
tmpfs                              5.0M     0  5.0M   0% /run/lock
/dev/sda2                          2.0G  104M  1.7G   6% /boot
tmpfs                              588M   12K  588M   1% /run/user/1000
wazuhadmin@wazuh-server:~$ free -h
               total        used        free      shared  buff/cache   available
Mem:           5.7Gi       483Mi       5.2Gi       1.5Mi       315Mi       5.3Gi
Swap:          4.0Gi          0B       4.0Gi
wazuhadmin@wazuh-server:~$ sudo whoami
[sudo] password for wazuhadmin:
root
wazuhadmin@wazuh-server:~$ sudo lvextend -l +100%FREE /dev/ubuntu-vg/ubuntu-lv
sudo resize2fs /dev/mapper/ubuntu--vg-ubuntu--lv
df -h  Size of logical volume ubuntu-vg/ubuntu-lv changed from <19.00 GiB (4863 extents) to <38.00 GiB (9727 extents).

  Logical volume ubuntu-vg/ubuntu-lv successfully resized.
wazuhadmin@wazuh-server:~$ df -h
Filesystem                         Size  Used Avail Use% Mounted on
tmpfs                              588M  1.6M  587M   1% /run
/dev/mapper/ubuntu--vg-ubuntu--lv   19G  6.6G   11G  38% /
tmpfs                              2.9G     0  2.9G   0% /dev/shm
tmpfs                              5.0M     0  5.0M   0% /run/lock
/dev/sda2                          2.0G  104M  1.7G   6% /boot
tmpfs                              588M   12K  588M   1% /run/user/1000
wazuhadmin@wazuh-server:~$ sudo resize2fs /dev/mapper/ubuntu--vg-ubuntu--lv
resize2fs 1.47.0 (5-Feb-2023)
Filesystem at /dev/mapper/ubuntu--vg-ubuntu--lv is mounted on /; on-line resizing required
old_desc_blocks = 3, new_desc_blocks = 5
The filesystem on /dev/mapper/ubuntu--vg-ubuntu--lv is now 9960448 (4k) blocks long.

wazuhadmin@wazuh-server:~$ df -h
Filesystem                         Size  Used Avail Use% Mounted on
tmpfs                              588M  1.6M  587M   1% /run
/dev/mapper/ubuntu--vg-ubuntu--lv   38G  6.6G   29G  19% /
tmpfs                              2.9G     0  2.9G   0% /dev/shm
tmpfs                              5.0M     0  5.0M   0% /run/lock
/dev/sda2                          2.0G  104M  1.7G   6% /boot
tmpfs                              588M   12K  588M   1% /run/user/1000
wazuhadmin@wazuh-server:~$
```

sudo resize2fs output and final df -h, confirming 38G total, 29G available.

6. Network Plan

Attribute	Value
Host machine	Dell PC (secondary), Windows, 8GB RAM, VMware Workstation Pro
Hypervisor network mode	Bridged
Assigned IP address	192.168.0.112/24 (DHCP-assigned, dynamic)
Network segment	192.168.0.0/24, the same physical home network segment as the router
Hostname	wazuh-server
Rationale	Bridged mode was chosen over NAT so wazuh-server is reachable from, and can reach, other lab VMs once they are also placed on the 192.168.0.x segment, which is required for those VMs to be enrolled as Wazuh agents in a later exercise.
Planned next step	Switch the primary laptop's Kali Linux, Metasploitable, and Windows 11 lab VMs from NAT to Bridged mode so they land on the same 192.168.0.x segment and can be reached by wazuh-server for agent enrollment.

7. Findings and Verification Summary

Check	Result	Status
Operating system	Ubuntu 24.04.5 LTS (Noble Numbat)	Supported
Hostname	wazuh-server	Confirmed
IP / network connectivity	192.168.0.112/24 via Bridged adapter; internet reachable (0% packet loss)	Confirmed
RAM	5.7GiB visible of 6GB allocated (below Wazuh's 8GB recommendation, documented compromise)	Confirmed, compromise noted
Disk	38GB usable, 29GB free after LVM correction (below Wazuh's 50GB recommendation, documented compromise)	Corrected, compromise noted
Administrative access	sudo whoami returns root	Confirmed
Network plan	Recorded in Section 6	Complete

8. Conclusion

The wazuh-server virtual machine is fully prepared for Wazuh installation. Its operating system, hostname, network connectivity, administrative access, and available RAM all meet requirements, and the disk allocation issue found during verification has been corrected. Two deliberate compromises against Wazuh's official small-deployment hardware recommendations remain, 6GB RAM against a recommended 8GB, and 38GB usable disk against a recommended 50GB, both accepted as documented tradeoffs necessitated by the host machine's hardware. With preparation complete, the environment is ready for the Wazuh installation covered in Day 17.